

Medical segmentation of anatomical structures with missing labels

Collaborators Imade Bouftini - Taha Boukaidi Laghzaoui - Ilyas Hlaili - Aicha Berquecha
 Supervisor Claire Br  cheteau
 Company Raidium

1 Introduction

Computed Tomography (CT) imaging is fundamental to clinical diagnosis and treatment planning, providing detailed cross-sectional views of internal anatomy. Manual segmentation of anatomical structures in CT scans is time-consuming and subject to inter-observer variability, necessitating Computer-Aided Detection (CAD) systems for automated organ delineation. This work addresses a CT multi-organ segmentation challenge featuring 800 partially annotated images (where visible organs may be unlabeled), 1200 unlabeled images, and 54 anatomical classes including tumors, evaluated on fully annotated test data using the Dice similarity coefficient.

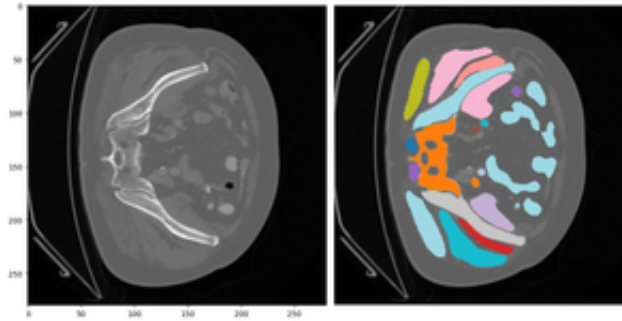


Figure 1: Example of CT scan segmentation — Source : challengedata.ens.fr

2 Problem Analysis

2.1 The Partial Annotation Problem

The competition dataset presents a particularly challenging variant of semi-supervised learning. The 800 annotated training images contain *partial annotations*: for any given image, only a subset of visible organs are labeled, while others, despite being present and visible, are assigned to the background class (label 0). This creates a fundamental problem for standard supervised learning.

In traditional semantic segmentation, the training objective assumes that background pixels (label 0) represent true background. However, under partial annotation:

$$\mathcal{Y}_{\text{background}} = \{y \mid y \in \text{Background}\} \cup \{y \mid y \in \text{Organs}_{\text{unlabeled}}\} \quad (1)$$

This induces contradictory supervision signals. Consider an organ o_i that appears labeled in image I_1 and unlabeled (marked as background) in image I_2 . Standard cross-entropy loss will penalize the model for correctly predicting o_i in I_2 , as the ground truth $y_0 = 1$ for those pixels. As a result, naive supervised learning strategies significantly degrade model performance.

2.2 Dataset Characteristics

1. **Scale and Complexity:** The dataset comprises 2000 2D CT slices at 256×256 resolution, spanning abdominal and thoracic regions, with 54 classes.
2. **Anatomical Variability:** CT slices exhibit substantial inter-patient anatomical variation and inter-slice positional variability. Organ boundaries are often ambiguous due to similar tissue densities, requiring the model to learn subtle intensity gradients.

3. **Evaluation Mismatch:** The test set contains full annotations for all 54 classes. Models must generalize to organs rarely or never seen during supervised training, making effective use of the 1200 unlabeled images critical.

2.3 Baseline Performance Gap

The competition organizers report that nnU-Net achieves 44% mean Dice on this task, while a provided watershed baseline scores 0.11%. This gap indicates that: Standard segmentation architectures can achieve reasonable performance if properly adapted, naive approaches fail due to their limited abilities and partial annotation problem, and there exists substantial room for improvement through semi-supervised learning.

3 Strategy

Our approach addresses the partial annotation problem through three core decisions:

3.1 Task Formulation

We explored two formulations for handling partial annotations: independent binary classification and multi-class segmentation with selective masking. Both approaches address the contradictory supervision problem but differ in how they structure the prediction space.

3.1.1 Binary Independent Classification

Standard multi-class segmentation uses a softmax output layer that forces competition between classes. For each pixel, the model outputs a probability distribution over 55 classes (54 organs + background), and training maximizes the probability of the correct class while suppressing others.

This fails under partial annotations. When an organ appears as background in image I_1 (because it's not annotated) but as a positive class in image I_2 , the model receives contradictory signals: predict organ o_i here, but don't predict it there. The softmax formulation exacerbates this because predicting one class automatically reduces probabilities for all others.

A natural reformulation treats the task as 54 independent binary classifiers, one per organ. Each classifier answers: "Is organ o_i present at this pixel?" The model outputs 54 channels with sigmoid activation:

$$p_i(x, y) = \sigma(z_i(x, y)), \quad i = 1, \dots, 54 \quad (2)$$

where z_i is the logit for organ i at pixel (x, y) , and σ is the sigmoid function. Each p_i is independent, meaning that predicting liver doesn't affect the probability of predicting spleen.

This eliminates contradictory supervision: if organ o_i is not annotated in an image, we skip loss computation for that classifier. However, independence introduces a critical limitation. Multiple organs can be predicted at the same pixel. At inference, converting 54 binary predictions to a single label requires argmax over confidence scores, which can produce false positives when multiple organs have similar high probabilities. More fundamentally, the approach discards anatomical constraints: organs are mutually exclusive in CT slices, yet the model never learns this structure. Each classifier operates in isolation, missing inter-class relationships that could improve boundary delineation between adjacent organs.

3.1.2 Multi-Class with Ignore Index Masking

Multi-class segmentation can handle partial annotations if we explicitly mark unlabeled regions as invalid rather than forcing them into the background class. The model outputs 55 logits per pixel and applies softmax for class probabilities, but crucially, loss computation ignores pixels assigned a special ignore index.

For each image, we construct a filtered label map:

$$\tilde{y}(x, y) = \begin{cases} y(x, y) & \text{if } y(x, y) \in \{0\} \cup O_j \\ -100 & \text{otherwise} \end{cases} \quad (3)$$

where O_j is the set of annotated organ IDs for image j . Background pixels (label 0) are always valid. Pixels belonging to unannotated organs receive the ignore index -100 , excluding them from gradient computation.

This formulation enforces mutual exclusivity, such that each pixel belongs to exactly one class. This assumption is consistent with anatomical reality (organs don't overlap in CT slices). The model learns inter-class relationships through the shared softmax output, potentially improving boundary delineation between adjacent organs.

Empirically, we found this multi-class approach outperforms independent binary classifiers by 5-8 percentage points in DICE score. The softmax competition, initially problematic under naive training, becomes beneficial when unlabeled regions are properly masked. The model learns that predicting liver should suppress kidney predictions at the same pixel, reducing false positives from overlapping binary predictions.

Given this performance advantage, all subsequent experiments and results use the multi-class formulation with ignore index masking exclusively.

3.2 Masked Loss Computation

Both formulations require masking to handle partial annotations, but they implement it differently.

Binary approach: Organ-level masking selectively computes loss per classifier. For each image, we identify annotated organs by checking which organ IDs appear in the label matrix. If organ o_i has no labeled pixels in image I_j , we skip loss computation for that classifier entirely. For annotated organs, we compute loss over all pixels. This includes both positive (organ present) and negative (organ absent). The loss for image j is:

$$\mathcal{L}_j = \frac{1}{|O_j|} \sum_{i \in O_j} (\mathcal{L}_{\text{BCE}}^i + \mathcal{L}_{\text{Dice}}^i) \quad (4)$$

where O_j is the set of annotated organs. Binary cross-entropy provides pixel-level supervision:

$$\mathcal{L}_{\text{BCE}}^i = -\frac{1}{N} \sum_{(x,y)} [t_i(x,y) \log p_i(x,y) + (1 - t_i(x,y)) \log(1 - p_i(x,y))] \quad (5)$$

Dice loss optimizes the segmentation metric directly:

$$\mathcal{L}_{\text{Dice}}^i = 1 - \frac{2 \sum_{(x,y)} p_i(x,y) t_i(x,y) + \epsilon}{\sum_{(x,y)} p_i(x,y) + \sum_{(x,y)} t_i(x,y) + \epsilon} \quad (6)$$

We weight both terms equally (1.0 + 1.0). BCE drives early learning while Dice refines boundaries.

Multi-class approach: Pixel-level masking via ignore index. `CrossEntropyLoss` with `ignore_index=-100` automatically skips pixels labeled `-100` during gradient computation. We combine two loss terms weighted 30% CE and 70% Dice:

$$\mathcal{L} = 0.3 \cdot \mathcal{L}_{\text{CE}} + 0.7 \cdot \mathcal{L}_{\text{Dice}} \quad (7)$$

The Dice component computes per-class overlap only for foreground classes ($c \geq 1$) on valid pixels (excluding ignore index). We apply class weighting: background at 0.1, all organs at 1.0, preventing trivial all-background predictions while focusing optimization on organ segmentation.

The ϵ term (10^{-5}) prevents division by zero. Dice loss handles class imbalance effectively—organs occupy small image fractions, but Dice weights each organ's overlap equally regardless of size.

3.3 Model Architectures

We tested multiple architectures including U-Net variants and DeepLab, which have proven effective for medical image segmentation. While we experimented with standard U-Net, Attention U-Net, and ResUnet, we focus here on the best-performing models.

U-Net foundation: U-Net is the canonical architecture for medical image segmentation. It follows an encoder-decoder structure with skip connections. The encoder progressively downsamples the input through convolutional blocks, extracting hierarchical features while reducing spatial resolution (256→128→64→32→16 pixels). The decoder mirrors this process, upsampling back to full resolution. Critically, skip connections directly concatenate encoder features with decoder features at each resolution level, allowing the decoder to recover fine spatial details lost during downsampling. This design works well for medical imaging because the encoder learns semantic features (what organs are present) while skip connections preserve spatial precision (where boundaries are located). All variants below build on this encoder-decoder-skip structure.

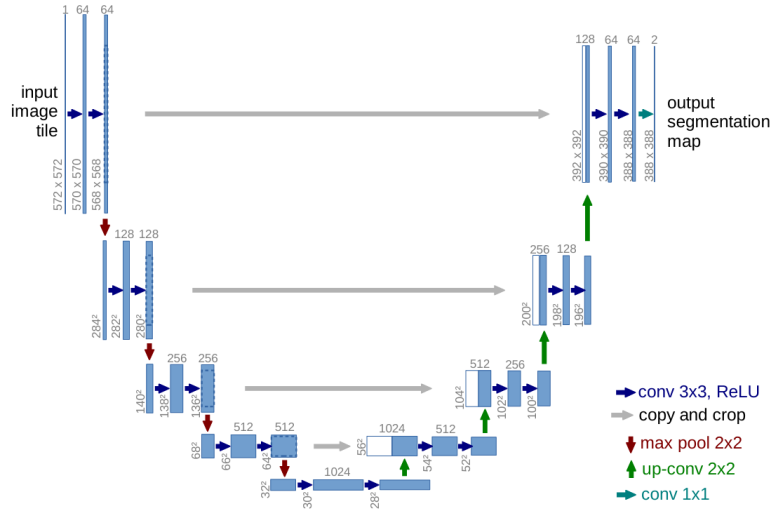


Figure 2: U-Net Architecture showing encoder-decoder structure with skip connections — Source: Ronneberger et al.

3.3.1 U-Net++

U-Net++ (Nested U-Net) redesigns skip connections to improve feature fusion between encoder and decoder. Standard U-Net directly concatenates encoder features with decoder features at each resolution level. This can be suboptimal when the semantic gap between shallow encoder features (low-level edges) and deep decoder features (high-level organ semantics) is large.

U-Net++ introduces nested skip pathways: instead of direct connections, it adds intermediate convolutional blocks between encoder and decoder at each level. For example, at the second level, encoder features pass through a convolutional block that receives both the encoder output and upsampled features from the level below. This creates dense connections where each decoder node aggregates features from all preceding encoder nodes at that resolution.

The nested architecture enables deep supervision. The model outputs predictions at multiple decoder depths during training. We supervise all intermediate outputs, computing loss at each level and averaging. This improves gradient flow to shallow layers and helps the model learn features at multiple semantic levels.

We combine U-Net++ with a ResNet-101 encoder pretrained on ImageNet, adapting to grayscale input. The decoder uses nested skip pathways with deep supervision. We weight the final output's loss at 0.7 and distribute 0.3 across intermediate outputs, emphasizing the deepest prediction while maintaining gradient flow.

The nested architecture increases parameters by 35% compared to standard U-Net, raising memory requirements. We use gradient checkpointing to trade computation for memory, enabling larger batch sizes. The deep supervision regularizes training effectively, reducing overfitting compared to standard U-Net despite the larger capacity.

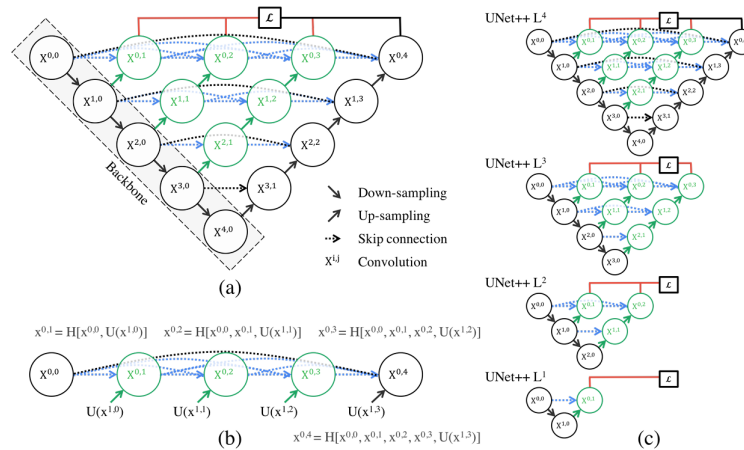


Figure 3: U-Net++ Architecture with nested skip pathways and deep supervision — Source: Zhou et al.

3.3.2 ResUNet

The ResUNet replaces the encoder with a pretrained ResNet backbone. We tested ResNet-34, ResNet-50, and ResNet-101, all pretrained on ImageNet.

Transfer learning makes sense here despite the domain gap. ResNet learns hierarchical features: edges and textures in early layers, object parts in middle layers, complex patterns in deep layers. These low-level features transfer well to medical imaging. Organ boundaries are still edges, tissue textures are still textures.

We adapt the first convolutional layer to accept single-channel grayscale input. Since the pretrained weights expect RGB (3 channels), we average the RGB weights into a single channel. This preserves the learned filters while matching our input format.

The decoder follows the U-Net design with skip connections from ResNet's intermediate layers. We add dropout (0.2) in decoder blocks to prevent overfitting, since the large pretrained encoder (25M+ parameters) can memorize the training set.

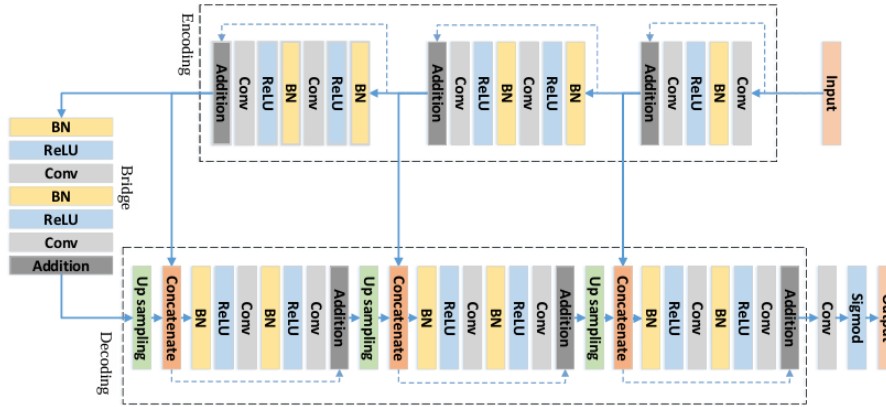


Figure 4: ResUNet Architecture — Source : Zhang et al.

In addition, we use differential learning rates. The encoder gets a lower learning rate (5×10^{-5}) because it's already trained and needs fine-tuning, not retraining. The decoder gets a higher learning rate (1×10^{-4}) because it's randomly initialized and needs to learn from scratch. This prevents catastrophic forgetting by destroying useful pretrained features through aggressive updates.

3.3.3 DeepLabV3+ with Atrous Convolution

DeepLabV3+ addresses a fundamental challenge in semantic segmentation: balancing spatial resolution with receptive field size. Standard convolution-pooling architectures reduce resolution aggressively, losing fine spatial details critical for precise boundary delineation.

The architecture uses atrous (dilated) convolution to expand receptive fields without reducing resolution. An atrous convolution with rate r inserts $r - 1$ zeros between filter weights, increasing the field of view by factor r while maintaining parameter count. The Atrous Spatial Pyramid Pooling (ASPP) module applies parallel atrous convolutions at multiple rates (6, 12, 18) plus global pooling, capturing multi-scale context in a single layer.

We use a ResNet-101 encoder pretrained on ImageNet, adapting the first layer from RGB to grayscale by averaging channel weights. The encoder output feeds into ASPP, then a lightweight decoder upsamples by $4 \times$ using low-level encoder features to recover spatial precision. This encoder-decoder design with atrous convolution achieves strong performance on organs with varying scales.

However, DeepLabV3+ requires careful hyperparameter tuning, since atrous rates must match the typical organ sizes in the dataset. We found the default rates (6, 12, 18) work reasonably but are not optimal for small organs like adrenal glands or blood vessels, which benefit from smaller receptive fields.

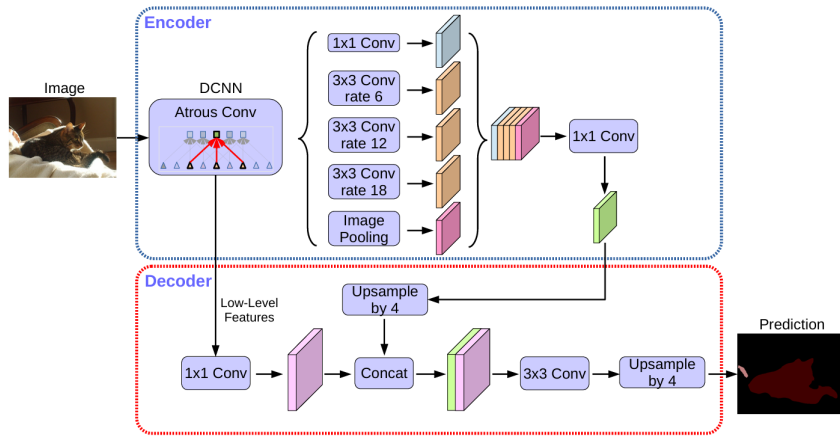


Figure 5: DeepLabV3+ Architecture showing ASPP module and decoder — Source: Chen et al.

3.4 Data Augmentation

Medical images have constraints that natural images don't. We use augmentations that increase data diversity while respecting anatomical structure.

Geometric augmentations include horizontal flips (50% probability, exploiting left-right anatomy symmetry), vertical flips (20% probability for patient orientation variation), and rotations up to $\pm 15^\circ$ (40% probability). We apply ShiftScaleRotate (40% probability) with 8% translation and 15% scaling to simulate patient positioning variations, using constant border padding to avoid introducing artifacts.

Intensity augmentations simulate imaging variation across scanners and acquisition protocols. RandomBrightnessContrast (40% probability) adjusts both by up to $\pm 15\%$, modeling different scanner calibrations. Gaussian noise (variance 10-30, 20% probability) mimics CT acquisition artifacts and detector noise, improving robustness to image quality variations.

We explicitly avoid elastic deformations and grid distortions, as these create anatomically impossible configurations that degrade performance. The key principle: augment within the space of realistic CT variations. All transformations preserve anatomical validity while increasing diversity.

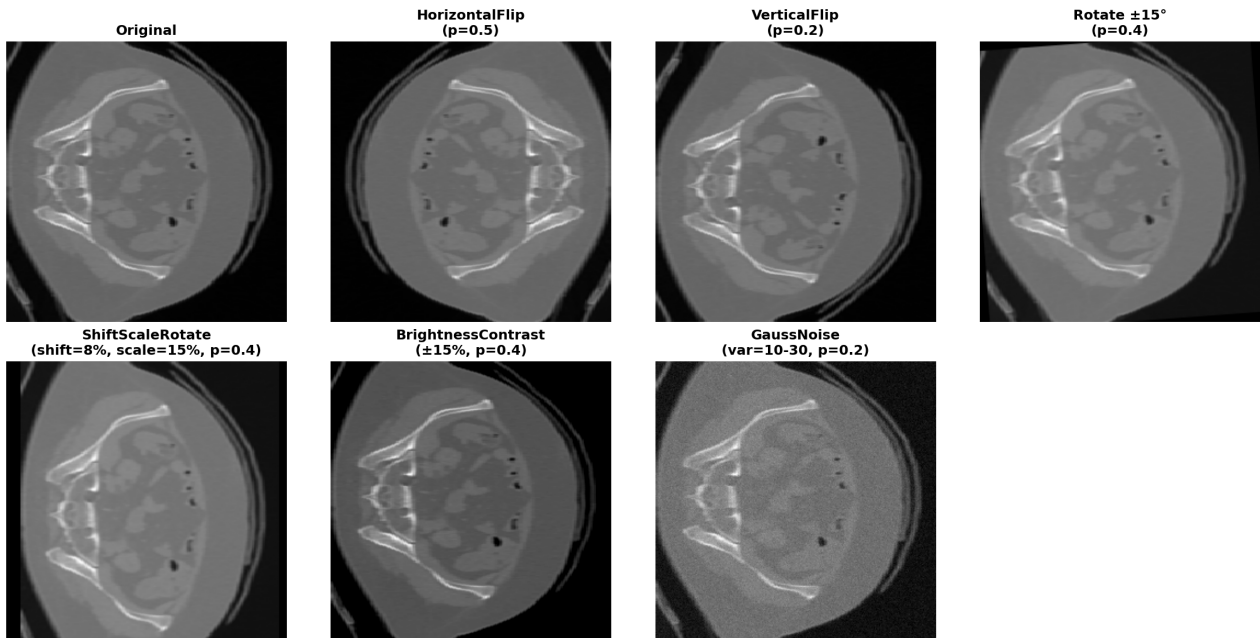


Figure 6: Individual augmentation effects applied separately to a CT scan. Each transformation (HorizontalFlip, VerticalFlip, Rotate, ShiftScaleRotate, BrightnessContrast, GaussNoise) is shown independently to demonstrate its impact on the input image.

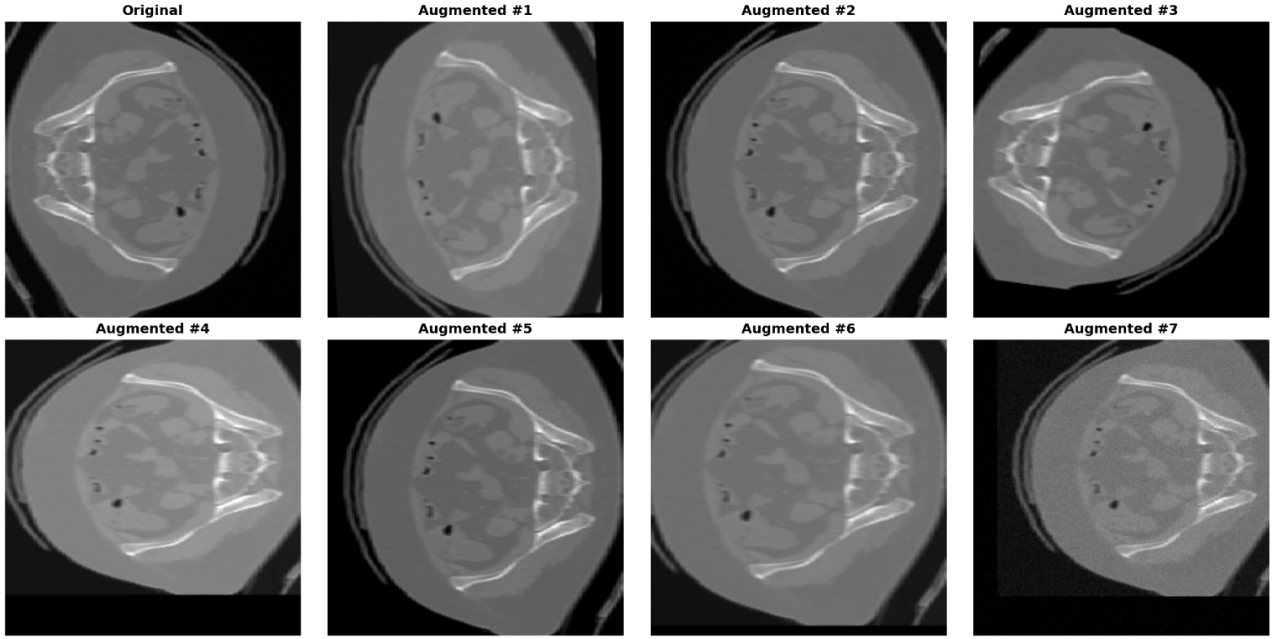


Figure 7: Combined augmentation pipeline applied multiple times. The first image shows the original CT scan, followed by seven randomly augmented versions demonstrating the diversity introduced during training. Each augmented sample results from the full transformation pipeline with stochastic application.

We rely on Albumentations for implementing these transformations. This library handles both image and mask augmentation simultaneously which is critical for segmentation tasks. When we flip or rotate the image, the corresponding segmentation mask must undergo the exact same transformation.

3.5 Evaluation Protocol

We compute Dice scores only for annotated organs, matching the competition protocol. This avoids penalizing predictions for unlabeled organs. The Dice coefficient for a single organ is:

$$\text{Dice}_{j,i} = \frac{2|P_{j,i} \cap T_{j,i}|}{|P_{j,i}| + |T_{j,i}|} \quad (8)$$

where $P_{j,i}$ and $T_{j,i}$ are the predicted and target binary masks for organ i in image j . The mean Dice score averages per-organ Dice within each image, then across images:

$$\text{Dice}_{\text{mean}} = \frac{1}{|I|} \sum_{j \in I} \frac{1}{|O_j|} \sum_{i \in O_j} \text{Dice}_{j,i} \quad (9)$$

where I is the set of validation images and O_j is the set of annotated organs in image j .

3.6 Technical details

We use AdamW optimizer with learning rate 8×10^{-4} and weight decay 10^{-4} . AdamW separates weight decay from gradient updates, providing better regularization than Adam's L2 penalty. The batch size is 10 with gradient accumulation over 2 steps, yielding an effective batch size of 20.

Learning rate scheduling follows polynomial decay: $\text{LR}(t) = \text{LR}_0 \times (1 - t/T)^{0.9}$ where t is the current epoch and T is the total epochs (150). This smooth decay avoids sharp drops and provides stable convergence. The minimum learning rate is 10^{-6} .

Early stopping monitors validation Dice score with patience of 15 epochs. Training terminates if no improvement occurs within this window, preventing overfitting while allowing the model sufficient time to escape plateaus.

Mixed precision training (FP16 forward/backward passes, FP32 master weights) reduces memory usage by 40% and accelerates training by $2\times$. This enables larger effective batch sizes on consumer GPUs. Gradient scaling prevents underflow in FP16 computations.

We train for 150 epochs with 20% validation split, using seed 42 for reproducibility. ImageNet normalization (mean=0.449, std=0.226) standardizes inputs. Checkpoints save every 10 epochs, with the best model selected by validation Dice score.

Experiments were conducted on Kaggle (NVIDIA Tesla P100). Training time: 2-3 hours per model. For dependencies, you can see requirements.txt.

4 Results

We evaluate both models on the validation set (114 images, 15% of annotated data) using the same evaluation protocol as the competition: mean DICE score computed only over annotated organs.

4.1 Training Dynamics

Each architecture exhibits distinct learning characteristics under the masked multi-class formulation with partial annotations.

4.1.1 ResUNet

The ResUNet demonstrates rapid initial convergence, reaching DICE 0.30 within 5 epochs and surpassing the 44% baseline by epoch 15. Training and validation losses track closely until epoch 20, after which a gap emerges due to augmentation applied exclusively during training. The validation DICE plateaus at approximately 0.60 around epoch 40, with minimal improvement thereafter despite continued loss reduction. This saturation indicates that the pretrained ResNet encoder has extracted maximal transferable features from ImageNet for this CT domain. The tight train-validation coupling suggests the model generalizes well without memorizing partial annotations.

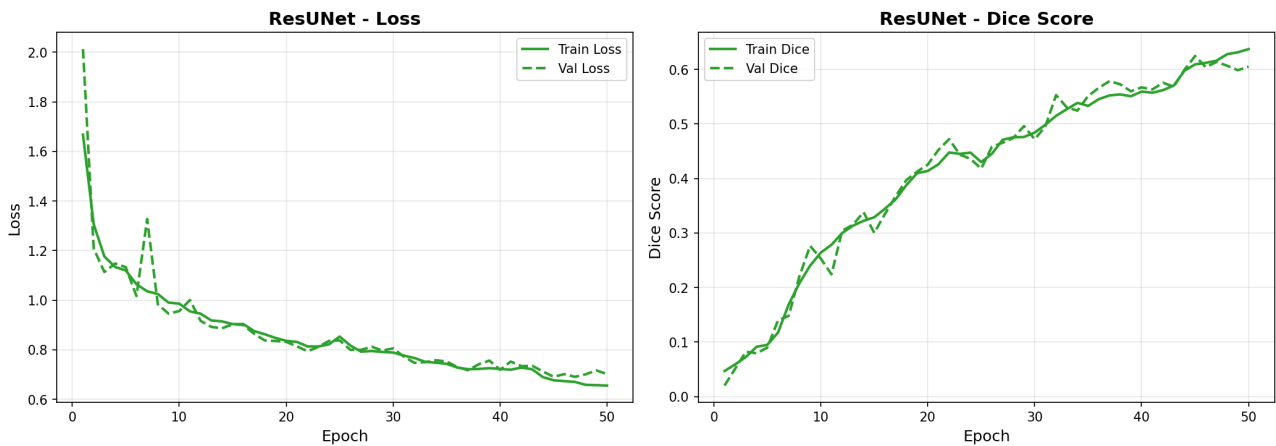


Figure 8: ResUNet training dynamics over 50 epochs. Left: Combined loss (0.3 CE + 0.7 Dice) decreases monotonically with train-validation gap emerging after epoch 20. Right: Validation DICE saturates at 0.60, indicating capacity limits despite pretrained features.

4.1.2 U-Net++

U-Net++ exhibits slower but more sustained learning compared to ResUNet. Initial convergence is gradual. The model requires 30 epochs to reach DICE 0.50, whereas ResUNet achieves this by epoch 12. However, performance continues improving beyond epoch 50, reaching 0.68 by epoch 100. The nested skip pathways and deep supervision appear to require more training iterations to optimize the hierarchical feature aggregation. Training loss remains consistently below validation loss throughout, with the gap widening after epoch 40. The validation DICE curve shows oscillations between epochs 60-80, suggesting the model navigates challenging optimization landscapes induced by the nested architecture's increased parameter count.

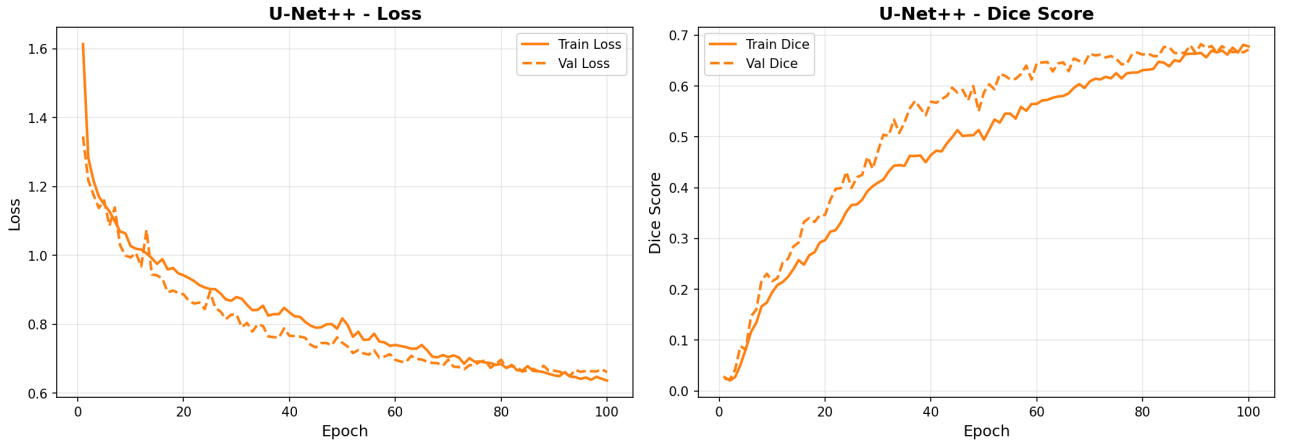


Figure 9: U-Net++ training dynamics over 100 epochs. Left: Loss convergence slower than ResUNet, with persistent train-validation gap. Right: Validation DICE exhibits late-stage improvements and oscillations, reaching 0.67 through extended training enabled by deep supervision regularization.

4.1.3 DeepLabV3+

DeepLabV3+ achieves the strongest performance (DICE 0.69) but displays training instability not observed in encoder-decoder architectures. The model reaches DICE 0.69 by epoch 20, then continues gradual improvement to 0.77 by epoch 30. Critically, validation DICE begins declining after epoch 30 while training DICE continues rising. This indicates overfitting despite weight decay and dropout. The validation loss curve shows irregular fluctuations post-epoch 15, potentially caused by atrous convolution's large receptive fields memorizing annotation patterns rather than learning anatomical structure. This architecture benefits from early stopping; peak validation performance occurs well before training completion.

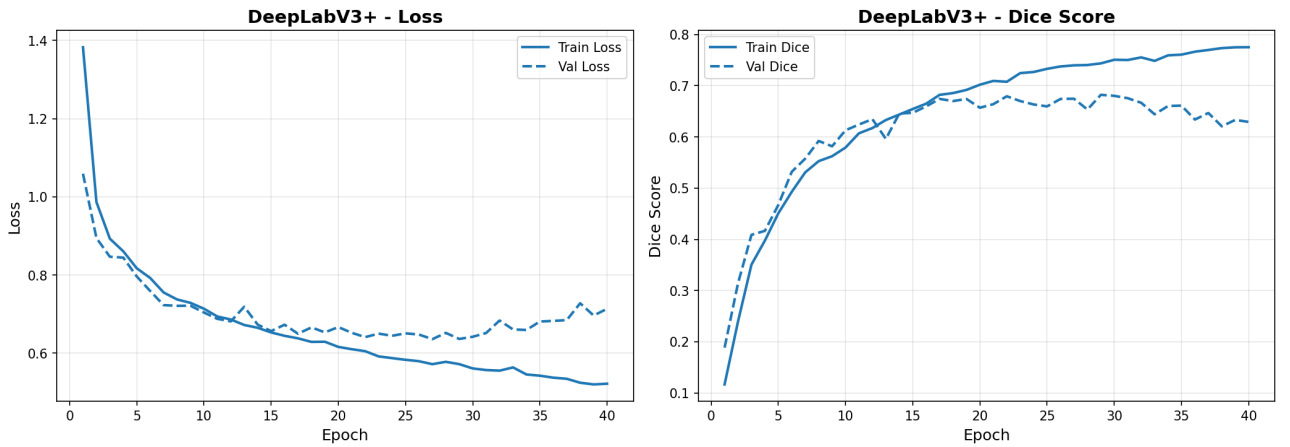


Figure 10: DeepLabV3+ training dynamics over 45 epochs. Left: Validation loss diverges from training loss and fluctuates after epoch 15, indicating optimization instability. Right: Validation DICE peaks at 0.77 around epoch 30, then degrades while training DICE continues increasing, indicating overfitting despite regularization.

4.1.4 Comparative Analysis

The three architectures reveal fundamental trade-offs. ResUNet converges fastest due to pretrained features but saturates earliest, limited by fixed ImageNet representations. U-Net++ requires extended training (100 vs 50 epochs) to reach comparable performance, trading sample efficiency for higher asymptotic accuracy through learnable nested aggregation. DeepLabV3+ achieves peak performance but demands careful regularization and early stopping, as its atrous spatial pyramid, while effective for multi-scale context, proves prone to overfitting under partial supervision where contradictory signals exist for unannotated regions.

The masked loss formulation provides stable gradients across all architectures. No catastrophic divergence occurs despite partial annotations. However, the augmentation-induced train-validation gap widens differently: ResUNet maintains 0.05 loss separation, U-Net++ reaches 0.10, and DeepLabV3+ exhibits irregular gaps exceeding

0.15. This suggests that architectural complexity (DeepLabV3+ $\geq$ U-Net++ $\geq$ ResUNet) increases susceptibility to distribution shift between augmented training data and clean validation data.

4.2 Qualitative Results

Figure 11 presents a representative thoracic CT slice comparing predictions from all three architectures. The ground truth annotates 9 organs: liver (green, dominant mass), lungs (purple, bilateral), heart chambers and vessels (blue, cyan), ribs and spine (yellow, magenta), and muscle structures.

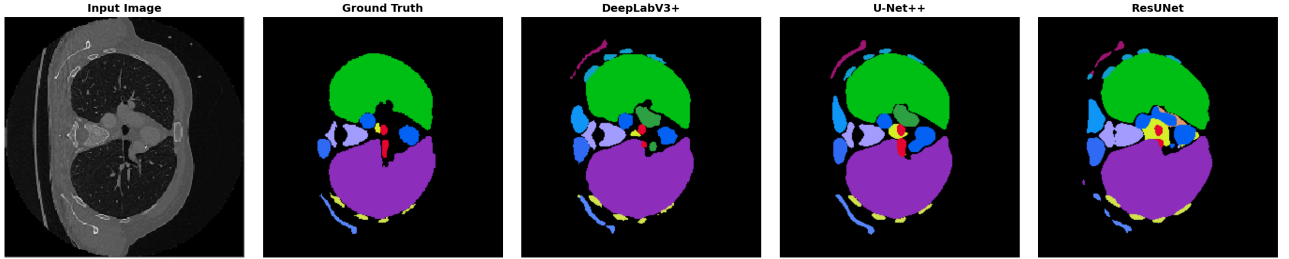


Figure 11: Qualitative comparison of model predictions on a thoracic CT slice. From left: input image, ground truth annotation, DeepLabV3+ prediction, U-Net++ prediction, ResUNet prediction. All models accurately segment major organs (liver, lungs) while exhibiting architecture-specific differences in small structure delineation.

All three models correctly segment the dominant anatomical structures. Liver boundaries align precisely with ground truth, and lung parenchyma segmentation shows high fidelity. However, subtle differences emerge in challenging regions:

Small organ detection: DeepLabV3+ captures the most complete set of small structures, particularly the yellow anatomical elements (ribs/vertebrae) along the lung boundary and spine. U-Net++ identifies these structures but with slightly reduced spatial extent. ResUNet misses several small yellow structures entirely, consistent with its lower DICE score and earlier saturation during training.

Boundary precision: DeepLabV3+ produces the sharpest organ boundaries, particularly visible in the heart-lung interface (cyan-purple boundary). The atrous spatial pyramid’s multi-scale context appears effective for resolving ambiguous tissue transitions. U-Net++ boundaries show minor irregularities, while ResUNet exhibits the smoothest but occasionally over-regularized contours.

Inter-organ gaps: All models leave small black gaps between organs rather than forcing continuous coverage. This behavior arises from the multi-class softmax formulation with a background class. When the model is uncertain between adjacent organs, it defaults to background (label 0). The gaps are smallest in DeepLabV3+, suggesting higher prediction confidence.

False positives: No model generates significant false positive organs absent from ground truth. The masked loss successfully prevents hallucination of structures not present in the annotation. Minor discrepancies (e.g., slight extension of blue cardiac structures) likely reflect ambiguity in manual annotation rather than model error.

The visual assessment confirms quantitative results: DeepLabV3+ achieves superior segmentation quality at the cost of training stability, U-Net++ provides a balanced trade-off, and ResUNet delivers reliable major organ segmentation but struggles with fine-grained structures. Importantly, all models demonstrate generalization beyond partial annotations, predicting anatomically plausible multi-organ segmentations despite training on images where many visible organs were unlabeled.

4.3 Semi-Supervised Exploration and Methodological Refinement

Following the supervised baseline results, we initiated a semi-supervised learning (SSL) phase to leverage the 1200 unlabeled images we still didn’t use. This exploration aimed to improve generalization to the fully annotated test data by exploiting the anatomical consistency present across the complete dataset.

An initial pilot using a Naive Mean Teacher (MT) implementation resulted in severe performance drop: validation Dice dropped from the supervised peak of 0.77 to the 0.30–0.40 range. This empirical finding confirmed the primary risk inherent to this dataset: the sheer volume of consistency loss signal ($\mathcal{L}_{\text{cons}}$) from unlabeled data rapidly overpowered the supervised loss ($\mathcal{L}_{\text{sup}}$), inducing label shift leading to forgetting of the learned organ boundaries.

4.3.1 Controlled Mean Teacher Implementation

To rigorously address the observed label shift, we upgraded the training protocol to a statistically robust and computationally controlled system:

- K-Fold Cross-Validation ($K = 5$): We implemented 5-fold CV over the entire 800 labeled images to maximize utilization of the limited labeled anchor set and provide statistically reliable performance estimates.
- 1:1 Batch Ratio Control: A custom dual-data-loader system guaranteed that every training step used an equal number of labeled and unlabeled samples (1:1 ratio). This design aimed to numerically anchor the supervised gradient signal against the exploration-driven consistency gradient, preventing label shift through balanced loss contributions.

Despite these methodological improvements, the controlled MT implementation yielded consistently poor results: upon completion of all K-fold experiments, the mean validation Dice converged to **0.40**, significantly below both the supervised baseline (0.77 validation Dice) and the final ensemble submission (0.59 test Dice). This stable underperformance indicates that the 1:1 volume control, while preventing catastrophic collapse, was insufficient to overcome the fundamental quality and distribution mismatch between the 800 labeled CT slices and the 1200 unlabeled images. This reveals a critical insight: the unlabeled data introduces systematic noise that cannot be mitigated solely through batch ratio engineering. Even with balanced gradient contributions, the model learns spurious patterns from low-confidence pseudo-labels on unlabeled images, degrading performance on anatomical structures that differ in prevalence or imaging characteristics between labeled and unlabeled subsets.

This negative result definitively establishes that volume-based control is necessary but not sufficient.

4.4 Summary of Results

The strongest individual supervised model, DeepLabV3+, achieved a peak validation Dice of 0.77 on the 114-image validation split. However, to maximize robustness on the fully annotated test set, we employed model ensembling for the final competition submission.

Ensemble Strategy: We combined predictions from four complementary models: DeepLabV3+ (epoch 30 checkpoint, 0.69 validation Dice, seed 42), DeepLabV3+ (epoch 30 checkpoint, 0.68 validation Dice, seed 123), U-Net++ with ResNet-101 encoder (epoch 100 checkpoint, 0.69 validation Dice, seed 42), and U-Net++ with ResNet-101 encoder (epoch 100 checkpoint, 0.68 validation Dice, seed 123). For each test image, we generated predictions from all three models using test-time augmentation (horizontal flip, original), yielding 8 predictions per image. Final segmentation masks were obtained through soft voting: averaging the class probability distributions across all predictions and selecting the class with maximum averaged probability at each pixel. This ensemble approach leverages the complementary strengths of each architecture. DeepLabV3+'s multi-scale context and U-Net++'s hierarchical feature aggregation. By averaging predictions, the ensemble also reduces overfitting exhibited by individual models.

The ensemble submission achieved a test set Dice of 0.59, establishing a robust supervised baseline that substantially exceeds the reported watershed baseline (0.11%) and surpasses the nnU-Net benchmark (0.44). The 18-point gap between validation Dice (0.77) and test Dice (0.59) arises primarily from evaluation protocol differences. Our validation split contains partial annotations (mean 12.3 organs per image), whereas the test set provides full annotations for all 54 classes. This forces the model to predict organs rarely seen during training, degrading per-organ Dice scores.

Our semi-supervised exploration using Mean Teacher, even with rigorous 1:1 batch ratio control and K-fold cross-validation, failed to improve upon the supervised baseline. Upon completion of all folds, the controlled MT implementation converged to a mean validation Dice of 0.40, falling short of both the individual DeepLabV3+ model (0.77 validation) and the final ensemble submission (0.59 test). This result confirms that volume-based control of unlabeled data is insufficient when the unlabeled set exhibits distribution shift or annotation quality mismatch relative to the labeled anchor set.

5 Conclusion

This work addresses multi-organ CT segmentation under partial annotations, where each training image labels only a subset of visible organs. We demonstrate that multi-class segmentation with ignore-index masking outperforms independent binary classification by 5–8 percentage points in Dice score, achieving competitive performance through proper handling of contradictory supervision signals.

5.1 Limitations and Future Directions

The primary limitation of this work is the failure to effectively leverage the 1200 unlabeled images. The negative MT results, however, provide clear guidance for possible future research.

The necessary pivot is from volume control to quality control. Rather than forcing the model to learn from all unlabeled data equally, we propose confidence-thresholded self-training as a possible future direction:

- Generate pseudo-labels on unlabeled images using the best supervised model (DeepLabV3+).
- Retain only pixels with maximum class probability $P_{\max} > 0.95$, discarding low-confidence predictions that introduce noise.
- Iteratively retrain on the union of ground-truth labels and high-confidence pseudo-labels, gradually expanding the training set with verified anatomical structures.

This approach prioritizes precision over coverage, adding unlabeled data only when the model demonstrates high certainty. By filtering out ambiguous regions where the unlabeled set likely differs from labeled data, confidence thresholding should provide stable improvements while avoiding the label shift that collapsed Mean Teacher.

Beyond quality-controlled SSL, several paths towards improvements are possible:

- **Supervised Fine-Tuning from MT Weights:** Despite the MT model's lower Dice score (0.40), its weights encode features learned from all 2000 images, potentially providing a more generalized initialization than ImageNet pretraining alone. A Phase 3 approach could initialize DeepLabV3+ with the MT Teacher's EMA weights and perform pure supervised fine-tuning (low learning rate $\sim 1 \times 10^{-5}$, supervised loss only) on the 800 labeled images. This treats MT as domain-specific pretraining, potentially recovering high Dice while retaining improved generalization.
- **Architecture Search:** Testing TransUNet, or Swin-UNETR architectures that have demonstrated state-of-the-art performance on medical segmentation benchmarks. nnU-Net's self-configuring preprocessing and training pipeline may be particularly well-suited to this dataset's heterogeneity.
- **Boundary-Aware Losses:** Incorporating boundary-weighted cross-entropy or Hausdorff distance losses to improve delineation of small organs (adrenal glands, blood vessels) where edge precision is critical.
- **Class-Balanced Weighting:** Dynamically adjusting per-class loss weights based on organ prevalence in the training set to address the long-tail distribution of anatomical structures.
- **Post-Processing Refinement:** Applying conditional random fields (CRFs) or learned post-processing networks to enforce anatomical constraints and spatial smoothness in final predictions.

5.2 Final Remarks

Despite the semi-supervised learning failure, this work establishes that standard segmentation architectures, when properly adapted through masked loss computation and ignore-index handling, achieve reasonable performance on partially annotated medical imaging tasks without requiring specialized SSL machinery. The supervised baseline demonstrates robustness to contradictory supervision signals and provides a strong foundation for future quality-controlled semi-supervised extensions.

The negative MT result, while disappointing, contributes valuable empirical evidence: in medical imaging datasets with potential distribution mismatch between labeled and unlabeled subsets, *selective data utilization through confidence thresholding is likely more effective than universal data inclusion through gradient balancing*. This insight should guide future semi-supervised medical segmentation research on partially annotated datasets.

Note on Code Submission

Although a Jupyter notebook (.ipynb) and an HTML export were requested, we chose to adopt a professional MLOps-oriented workflow using a WSL/VSCode environment rather than an academic notebook format. This decision was motivated by our desire to familiarize ourselves with industry standards in machine learning and deep learning engineering.

This setup allowed us to separate the different stages of the pipeline (preprocessing, training, and evaluation) into modular Python scripts, while organizing checkpoints, visualizations, and data into dedicated folders. Through this structure, we learned best practices for reproducibility, experiment tracking, and scalable project organization.

For submission purposes, the provided .py files are self-contained and complete: the previously modular scripts have been merged into clearly delimited sections, each accompanied by detailed comments explaining its functionality. The overall structure follows standard machine learning engineering practices and facilitates reproducibility, understanding, and future extension of the work.

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